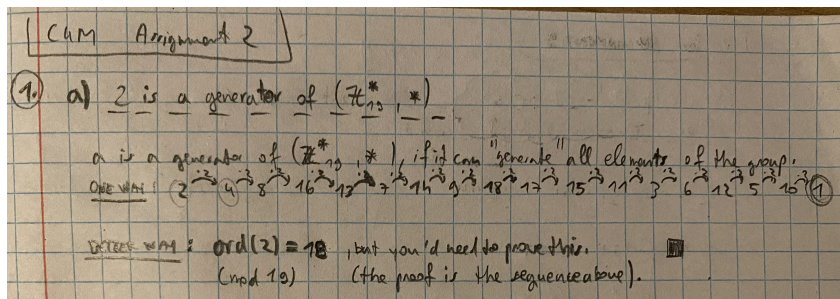


M4C 2025 - Assignment 2

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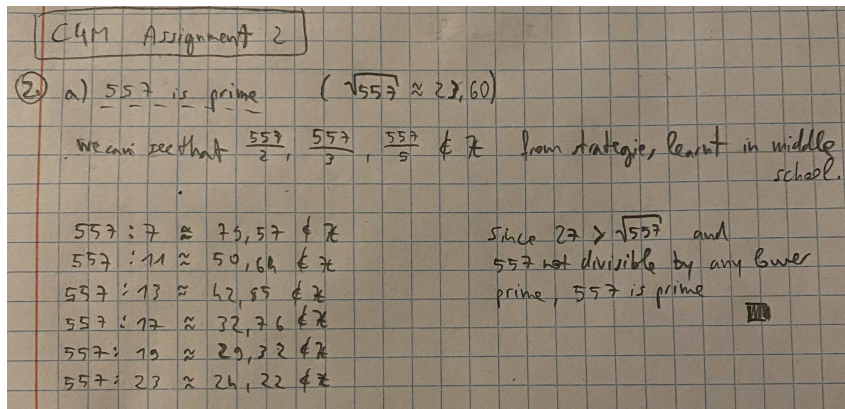
November 30, 2025

Problem 1



- (a)
- (b) Because 2 is a generator of the group $\mathbb{Z}_{19}$, every element is a power of 2 (mod 19). The elements in our group that have order 18 have to be elements that are powers of 2 and the power is coprime with 18. What I mean by that is if for example we want to see if 3 has order 18, we have to check how we get 3 from generator 2: $3 \equiv 2^{13}$ which means we can calculate powers of 3 with powers of 2, for example: $3^4 \equiv 2^{13 \cdot 4}$. To get the order of 3, we need to find the smallest number k that will result in: $3^k \equiv 1$ which can be translated into: we need to find the smallest number k so that $18 \mid 13 \cdot k$. Because 13 and 18 are coprime, our only option is $k = 18$.
- Following the same principle we can conclude that these elements (besides 2) have order 18: 3, 10, 13, 14, 15.
- (c) If a group G has a prime order p (p number of elements), then the order of every element in the group has to be either 1 or the prime number p . This is because the corollary after the Lagrange's theorem states, that the order of any element in the group has to divide the number of elements in that group. Since p is prime (it's only divisible by 1 and p), all of the orders are either 1 or p . Since 2 is the lowest prime number, we have 2 or more elements. If all the elements were of order 1, they would all be equivalent to the identity element, which would mean that we only had 1 element (contradictory with the fact that we have 2 or more elements). Therefore at least one element has order p , which means that it generates the whole group, which means the group is cyclic.

Problem 2



(a)

Deeper understanding: If we want to check that p is prime, we can only check the first few prime numbers that are smaller (or equal) than $\sqrt{p}$. The reason for this is that if p is not prime (and $\sqrt{p} \notin \mathbb{Z}$) then p has at least 2 divisors $a, b \in \{2, 3, \dots, p-1\}$ so that $p = ab$. Both a and b cannot be larger than $\sqrt{p}$, since they would not multiply to p .

- (b) Fermat's little theorem can't really help us in this case (or at least I don't see a useful connection). It tells us that $x^{556} \equiv 1 \pmod{557}$ for every integer x that isn't divisible by 557.
- (c) If I understood the question correctly then x is an element of $2, 3, \dots, 556$. We removed 0 and 1 because $0^k = 0 \neq 321 \pmod{n}$ and $1^k = 1 \neq 321 \pmod{n}$ for $n > 1$ which is our case.
- (d) while looking through youtube to find a good explanation for why we can use extended euclidean algorithm for finding the inverse, I found that this one explains it well. I used it to solve this problem: [link](#)

EXTENDED: $556 = 4 \cdot 127 + 46$
 $x=0, y=1$ $x=1, y=0$ $x=-4, y=-1$

$127 = 2 \cdot 46 + 35$
 $x=1, y=0$ $x=-4, y=-1$ $x=9, y=2$

$46 = 1 \cdot 35 + 11$
 $x=9, y=2$ $x=-17, y=-3$

$35 = 1 \cdot 11 + 24$
 $x=-17, y=-3$ $x=22, y=5$

$11 = 1 \cdot 24 + 3$
 $x=22, y=5$ $x=-35, y=-8$

$24 = 4 \cdot 3 + 2$
 $x=-35, y=-8$ $x=62, y=32$

$3 = 1 \cdot 2 + 1$
 $x=62, y=32$ $x=-197, y=-45$

$127x - 556y = 1$
 $x = 127^{-1} \pmod{556}$
 $127x = 1 \pmod{556}$
 $127x = 1 + 556y$
 $127x - 556y = 1$

$x = -197$
 $y = -45$
 $556 - 197 = 359$

The result was: $-197 \equiv 556 - 197 \equiv 359 \pmod{556}$

Observation: This method looks promising for implementation into python but too error prone for hand calculations.

- (e) Please correct me if I'm wrong, but because 557 is prime, we can translate our problem into a cyclic group problem where we look at group $\text{mathbb{Z}}_p^*$. Here we also learnt that $p \in \mathbb{P} \implies \mathbb{Z}_p^* \cong (\mathbb{Z}_{p-1}, +)$. So when the base is in mod 557, we can regard the exponent as being in mod 556. So for that reason $(x^{127})^{357} = (x^{127})^{127^{-1}} = x^{127 \cdot 127^{-1}} = x^1$.

From your instructions I speculated that we can maybe generalize this notion for when 557 is not prime. So for $m \in \mathbb{N}$ we could generally say that $\mathbb{Z}_m^* \equiv (\mathbb{Z}_{\phi(m)}, +)$. I proposed this question to duck.ai which said that we cannot generally speculate this because the structures are generally different: $\mathbb{Z}_m^*$ is not generally cyclic and $(\mathbb{Z}_n, +)$ is always cyclic for $n \in \mathbb{N}$. They do, however, have the same cardinality (although I don't know if this could be useful).

- (f) I answered this question with (e). The final solution to the congruence is then of course: $321^t = 321^{359}$.
- (g) since I didn't really understand the way we did the square-and-multiply algorithm in class, I once again found a short explanation video which I used as a reference: [link](#) Here are my computations:

9) $321^{359} = ? \pmod{557}$

1. step: Find binary of exponent: $359_{10} = 1\ 0\ 1\ 1\ 0\ 0\ 1\ 1\ 1$
256 128 64 32 16 8 4 2 1

2. step: How many digits? (9)

3. step: Table

1	2	3	4	5	6	7	8	9
1	0	1	1	0	0	1	1	1
321	557	123	483	463	487	400	144	106

4. step: If binary number is 0, we need to square (the previous)
 If binary number is 1, we need to square and multiply

computations: $321^2 \pmod{557} = 553$

$400^2 \pmod{557} = 141$

$141 \times 321 \pmod{557} = 144$

$144^2 \pmod{557} = 123$

$123 \times 321 \pmod{557} = 106$

$553^2 \pmod{557} = 16$

$16 \times 321 \pmod{557} = 123$

$123^2 \pmod{557} = 90$

$90 \times 321 \pmod{557} = 483$

$483^2 \pmod{557} = 463$

$463^2 \pmod{557} = 487$

$487^2 \pmod{557} = 206$

$206 \times 321 \pmod{557} = 400$

If my computations are correct (from which I'm sure there's a large probability of human error), the solution is: $321^{359} \pmod{557} = 106$.

When the opportunity arises, I will look into why the square-and-multiply method works. For now I will be okay with the result.

Problem 3

- (a) We have a general formula $a + b\sqrt{5}$. This means that we can derive a and b from our examples: $-3\sqrt{5} - 11 \Rightarrow a = -11, b = -3$. Let's check the square: $(-3\sqrt{5} - 11)^2 = 45 + 2 \cdot 33\sqrt{5} + 121 = 66\sqrt{5} + 166 \Rightarrow a = 166, b = 66$.

For our particular examples we can choose: $a_1 = 2, b_1 = 0 \Rightarrow ex_1 = 2$,
 $a_2 = 0, b_2 = 2 \Rightarrow ex_2 = 2\sqrt{5}, a_3 = -1, b = 1 \Rightarrow ex_3 = \sqrt{5} - 1$.

From these definitions we can easily check that the sums and products still hold the same form. For example if we calculate the sums and products of ex_1 and ex_2 : $ex_1 + ex_2 = 2\sqrt{5} + 2$ and $ex_1 * ex_2 = 4\sqrt{5}$. The sums and products of the other two pairs is done in the same fashion.

We can now make sure that the squares also hold shape:

$$ex_1^2 = 4 \Rightarrow a = 4, b = 0; ex_2^2 = 20 \Rightarrow a = 20, b = 0;$$

$$ex_3^2 = 6 - 2\sqrt{5} \Rightarrow a = 6, b = -2.$$

3. $T = \{a + b\sqrt{5} \mid a, b \in \mathbb{R}\}$ is a group

1. T is closed for $+$: Let $a_1, b_1, a_2, b_2 \in \mathbb{R} \Rightarrow a_1 + b_1\sqrt{5}, a_2 + b_2\sqrt{5} \in T$

$$(a_1 + b_1\sqrt{5}) + (a_2 + b_2\sqrt{5}) = (a_1 + a_2) + (b_1\sqrt{5} + b_2\sqrt{5}) =$$

$\uparrow$
assoc in $\mathbb{R}$

$$= (a_1 + a_2) + (b_1 + b_2)\sqrt{5} \in T$$

$\uparrow$
Dist in $\mathbb{R}$

2. $+$ is associative: Let $a_1, b_1, a_2, b_2 \in \mathbb{R} \Rightarrow a_1 + b_1\sqrt{5} \in T$

$$((a_1 + b_1\sqrt{5}) + (a_2 + b_2\sqrt{5})) + (a_3 + b_3\sqrt{5}) =$$

$$= ((a_1 + a_2) + (b_1 + b_2)\sqrt{5}) + (a_3 + b_3\sqrt{5}) =$$

$\uparrow$
assoc in $\mathbb{R}$

$$= (a_1 + a_2 + a_3) + (b_1 + b_2 + b_3)\sqrt{5} =$$

$\uparrow$
Dist in $\mathbb{R}$

$$= (a_1 + b_1\sqrt{5}) + ((a_2 + a_3) + (b_2 + b_3)\sqrt{5}) \stackrel{1.}{=} (a_1 + b_1\sqrt{5}) + (a_2 + b_2\sqrt{5}) + (a_3 + b_3\sqrt{5})$$

3. identity element $\neq 0$: $a + b\sqrt{5} + 0 \stackrel{\text{assoc}}{=} a + (b\sqrt{5} + 0) \stackrel{\text{identity in } \mathbb{R}}{=} a + b\sqrt{5}$
 $0 + a + b\sqrt{5} = (0 + a) + b\sqrt{5} \stackrel{\text{identity in } \mathbb{R}}{=} a + b\sqrt{5}$

4. Each element is invertible: $a, b \in \mathbb{R} \Rightarrow a + b\sqrt{5} \in T$

$-a - b\sqrt{5}$ is an inverse of $(a + b\sqrt{5})$

We can see that $-a - b\sqrt{5} \in T$ because $-a, -b \in \mathbb{R}$.
 so we need to check: $(a + b\sqrt{5}) + (-a - b\sqrt{5}) \stackrel{\text{assoc, dist}}{=} (a + (-a)) + (b + (-b))\sqrt{5} = 0 + 0\sqrt{5} = 0$

I will not check that $-a - b\sqrt{5}$ is also the left inverse, because we know that $(\mathbb{R}, +)$ is commutative and associative.

(b)

- (c) We can check that $(T, *)$ is closed and associative like we checked for addition. We can also check that the identity element for this group is 1.

- (d) Generally (almost for all) an element of $(T, *)$ doesn't have an inverse. Here is a counter example:

$$(2 + 3\sqrt{5}) * (x + y\sqrt{5}) = 1$$

$$(2 * x + 3 * x * \sqrt{5} + 2 * y\sqrt{5} + 15 * y) = 1$$

$$((2x + 15y - 1) + (3x + 2y)\sqrt{5}) = 0$$

but because the first parentheses belong to integers and the second parentheses doesn't, both have to be 0. If we solve the two equations we get: $x = \frac{-2}{41}, y = \frac{3}{41} \notin \mathbb{Z}$ which means that the element $2 + 3\sqrt{5} \in (T, +, *)$ doesn't have an inverse in the group. Which further means that $(T, +, *)$ is not a field, only a ring.